Outline

- Inter-observational Distances
- Partition-based Clustering
- Hierarchical Clustering
- Diagnostics and Optimizing Number of Clusters
- Density-based Clustering

Optimal Cluster Counts

But… how do we choose the optimal number of clusters?

There is no single compelling way to choose clusters, but we will explore two types of methods:

- Direct Methods that involve optimizing a particular criterion
  - Silhouette method
  - Elbow method
- Testing Methods that evaluate evidence against a null hypothesis
  - Gap Statistic

Elbow Method

As previously, let $W(C_k)$ be the within-cluster sum of squared distances between all pairs for cluster k for a particular clustering, and let

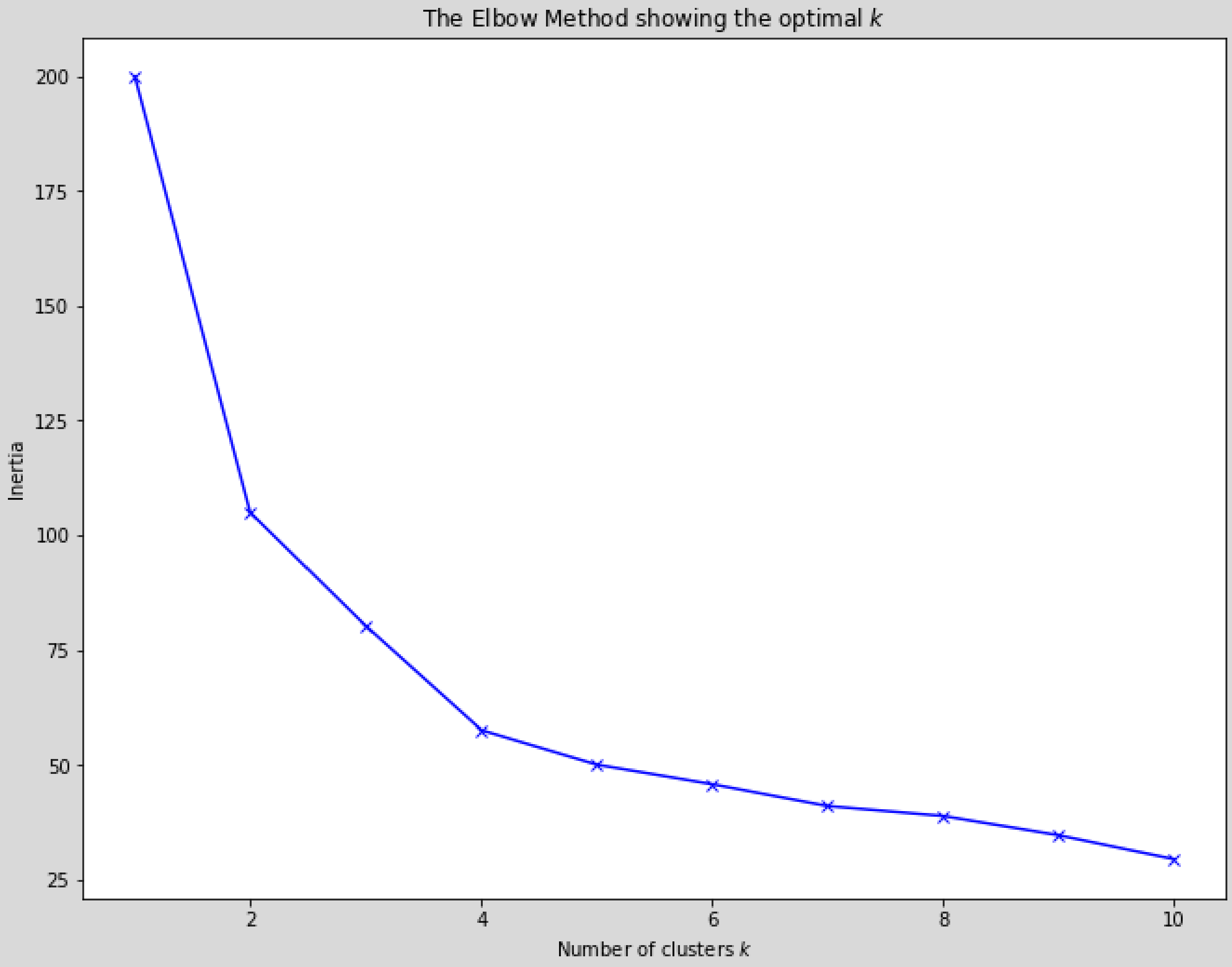
$$T_K = \sum_{k=1}^{K} W(C_k)$$

be the total within-cluster variation for the clustering of K clusters.

1. For the particular clustering method, let K vary over a range of values (e.g., 1 to 10)
2. Compute T_K for each K
3. Plot T_K against K and look for a clear bend in the graph

The value where the bend occurs is considered the correct number of clusters

Elbow Method



Average Silhouette Method

Similar to the elbow method:

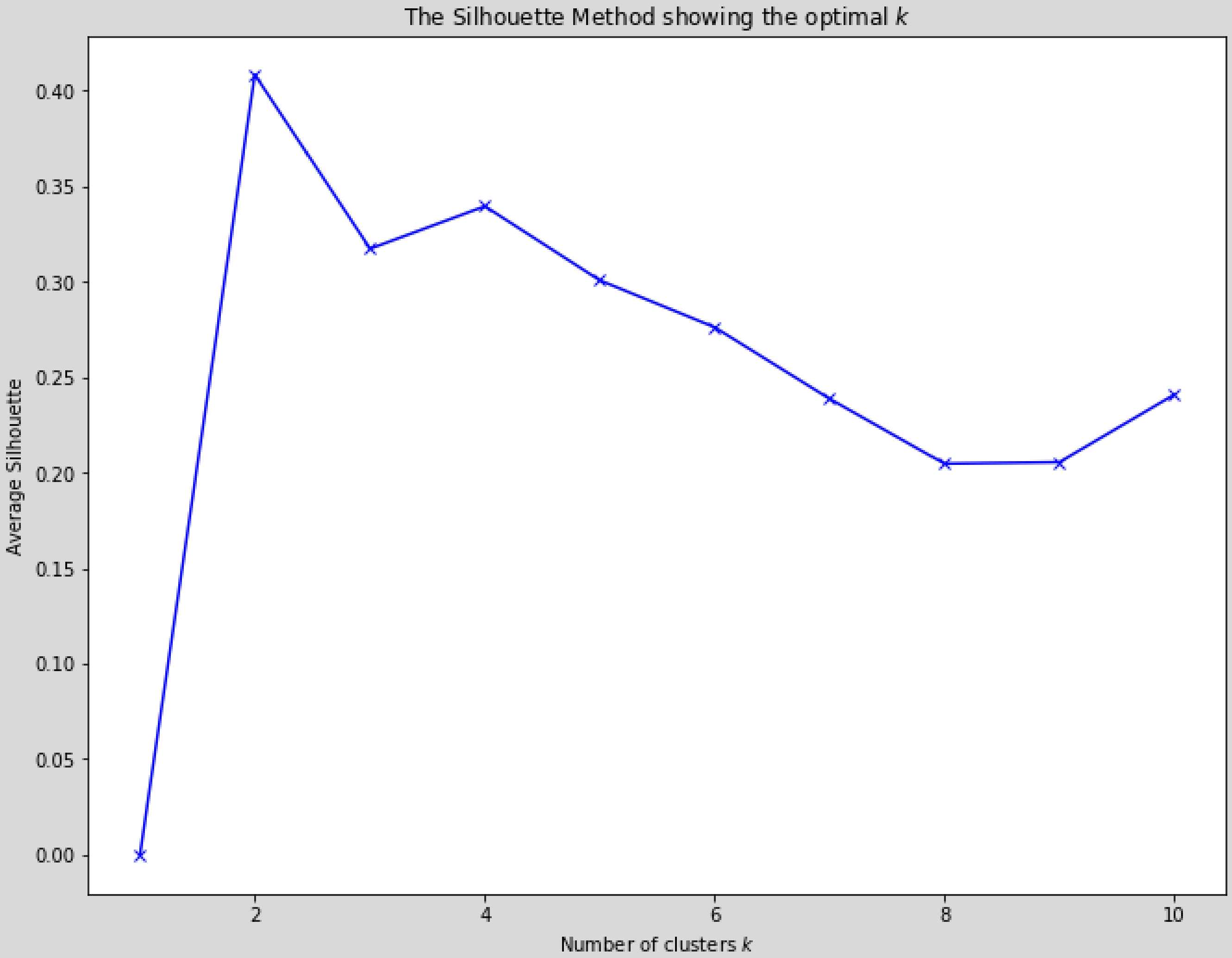
1. For the particular clustering method, let K vary over a range of values (e.g., 1 to 10)
2. For each K, calculate the average silhouette across all observations

$$S_K = \frac{1}{n}\sum_{i=1}^{n} s_i$$

3. Plot S_K against K

The value of K where S_K is maximized is considered the appropriate number of clusters.

Average Silhouette Method



Comments

- The elbow method suggests four clusters for K-Means
- Average silhouette method suggests two
- Computation for each method are different enough that inconsistent results are not uncommon
- Both approaches measure global clustering characteristics only, and are informal (and somewhat ad hoc approaches)

Gap Statistic

- Idea: For a particular choice of K clusters, compare the total within-cluster variation to the expected within-cluster under the assumption that the data have no obvious clustering (i.e., randomly distributed)

- The Gap Statistic in essence detects whether the data clustered into K groups is significantly better than if they were generated at random

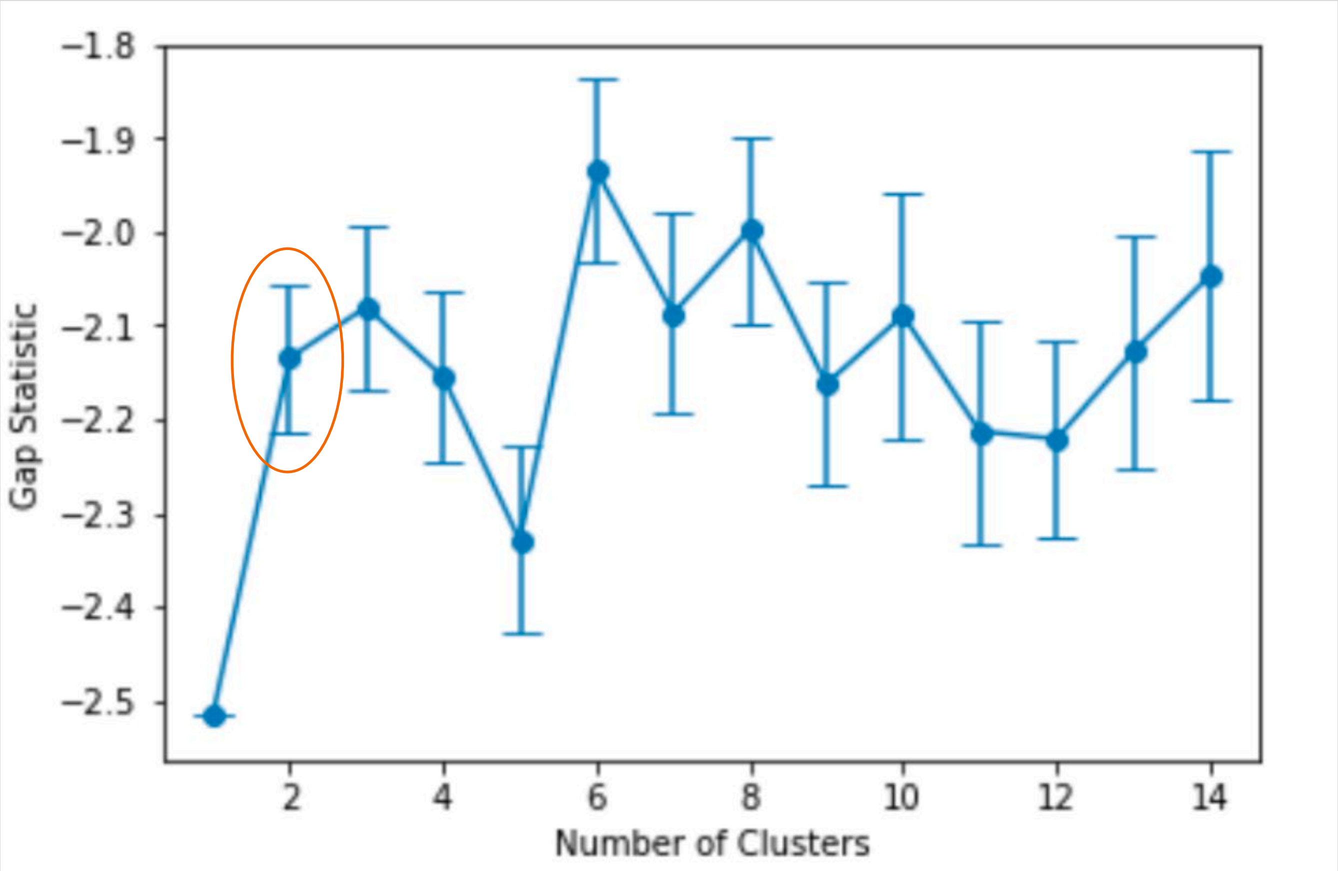
Gap Statistic Algorithm

1. Cluster the data at varying number of total clusters K. Let T_K be the total within-cluster sum of squared distances
2. Generate B reference data sets of size n, with the simulated values of variable j uniformly generated over the range of the observed variable x_j. Typically, $B = 500$.
3. For each generated data set $b = 1, \ldots, B$, perform the clustering for each K. Compute the total within-cluster sum of squared distances, $T_k^{(b)}$.
4. Compute the Gap Statistic: $Gap(K) = (\frac{1}{B}\sum_{b=1}^{B} \log\left(T_K^{(b)}\right)) - \log(\mathrm{T_K})$
5. Let $\bar{w} = \frac{1}{B}\sum_{b=1}^{B} \log\left(T_K^{(b)}\right)$. Compute the standard deviation $sd(K) = \sqrt{\frac{1}{B}\sum_{b=1}^{B}\left(\log\left(T_K^{(b)}\right) - \bar{w}\right)^2}$. Define $s_K = sd(K)\sqrt{1 + {}^{1}/_{B}}$.
6. Finally, **choose the number of clusters as the smallest *K* such that**
$$Gap(K) \geq Gap(K+1) - s_{k+1}$$

Gap Statistic Algorithm

	n_clusters	gap_value	gap*	ref_dispersion_std	diff	diff*
0	1.0	-2.518049	-183.840402	1.082720	-0.327729	1.418971
1	2.0	-2.108007	-92.168043	1.061159	0.042001	48.885993
2	3.0	-2.063404	-70.175014	0.892529	-0.011188	29.238100
3	4.0	-1.953498	-49.457242	0.798298	0.164137	42.875510
4	5.0	-2.013890	-45.934873	0.734503	0.098662	35.904791

Gap Statistic Algorithm



Gap Statistic Algorithm

- K-Means clustering is optimized for $K = 2$ based on the Gap Statistic
- This is a principled approach to choosing cluster sizes, though clearly different conclusions can be reached based on different clustering approaches
- The Gap Statistic is generally understood to be conservative—tends to err on the side of picking a smaller number of clusters

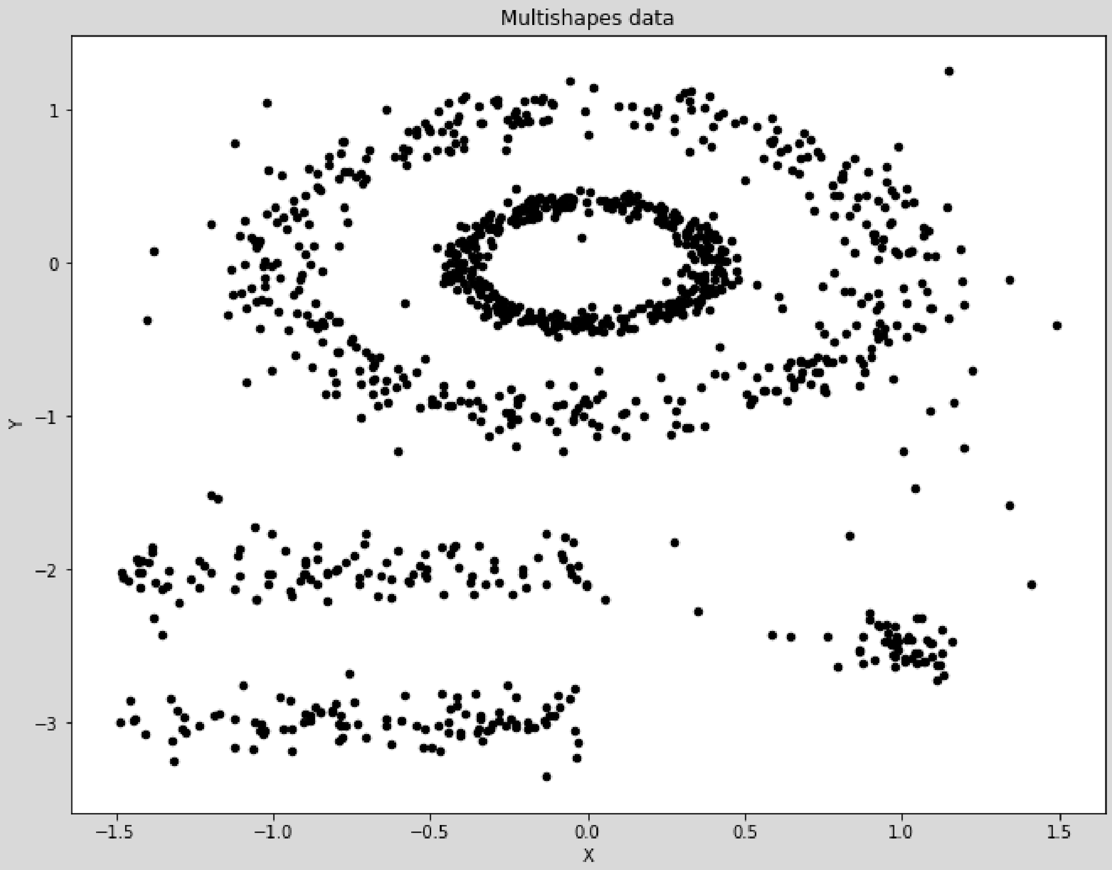
- There's an alternative form of the gap statistic which replaces the $\log(T_K)$ and $\log(T_K^{(b)})$ terms in the formula with

$$Gap^*(K) == (\frac{1}{B}\sum_{b=1}^{B} T_K^{(b)}) - \mathrm{T_K}$$

This turns out to be less conservative.

More Clustering

- Imagine trying to cluster something like this



What happens when we try K-means?

More Clustering

- K-Means struggles with this data set.